# Practical 4

## Program –

```
const int LDR_PIN = A0;
const int LED_PIN = 8;

String command = "";

void setup() {
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Serial.begin(9600);

  Serial.println("================================");
  Serial.println("   CPS MONITORING AND CONTROL UI");
  Serial.println("================================");
  Serial.println("Commands:");
  Serial.println("ON     - Turn LED ON");
  Serial.println("OFF    - Turn LED OFF");
  Serial.println("STATUS - View system status");
  Serial.println("HELP   - Display commands");
  Serial.println("--------------------------------");
}

void loop() {

  // Read sensor data
  int lightValue = analogRead(LDR_PIN);

  // Monitor the sensor continuously
  Serial.print("Light Sensor Value: ");
  Serial.print(lightValue);

  Serial.print(" | LED Status: ");
  Serial.println(digitalRead(LED_PIN) ? "ON" : "OFF");

  // Check if a command is received
  if (Serial.available() > 0) {

    command = Serial.readStringUntil('\n');
    command.trim();
    command.toUpperCase();

    // Turn LED ON
    if (command == "ON") {
      digitalWrite(LED_PIN, HIGH);
      Serial.println("LED turned ON.");
```

```
    }

    // Turn LED OFF
    else if (command == "OFF") {
      digitalWrite(LED_PIN, LOW);
      Serial.println("LED turned OFF.");
    }

    // Display system status
    else if (command == "STATUS") {
      Serial.println("--------- SYSTEM STATUS ---------");
      Serial.print("Light Sensor Value: ");
      Serial.println(lightValue);

      Serial.print("LED Status: ");
      Serial.println(digitalRead(LED_PIN) ? "ON" : "OFF");

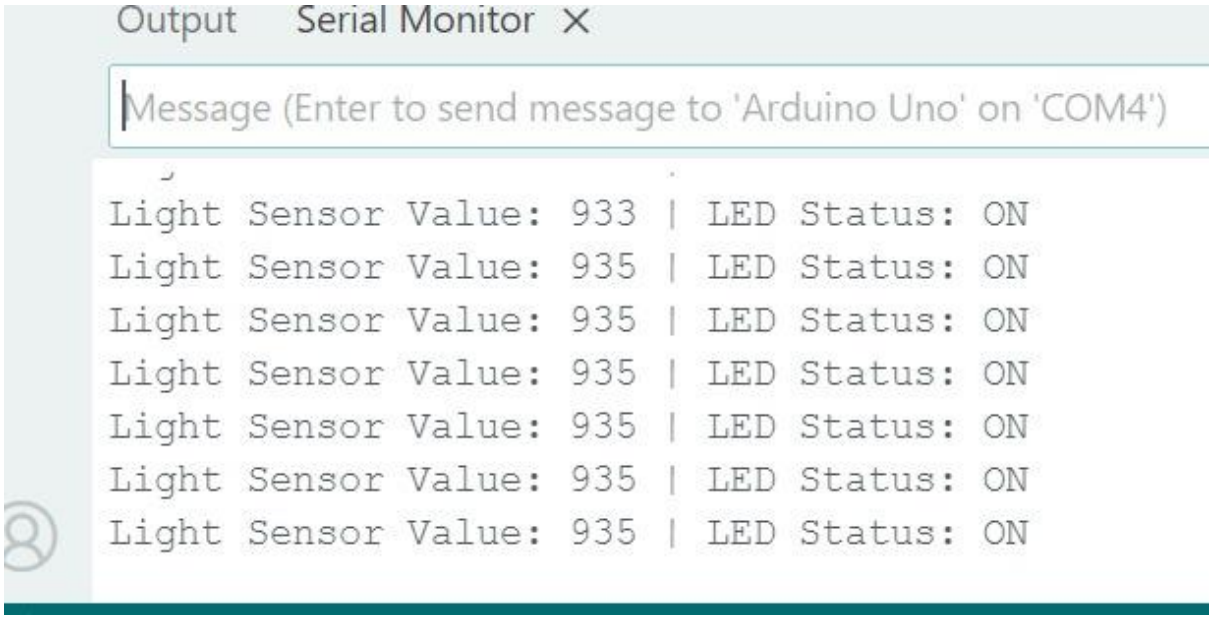
      Serial.println("--------------------------------");
    }

    // Display available commands
    else if (command == "HELP") {
      Serial.println("Available Commands:");
      Serial.println("ON     - Turn LED ON");
      Serial.println("OFF    - Turn LED OFF");
      Serial.println("STATUS - View system status");
      Serial.println("HELP   - Display commands");
    }

    // Invalid command
    else {
      Serial.println("Invalid command!");
      Serial.println("Type HELP to view available commands.");
    }
  }

  delay(1000);
}
```

**Output**


Output
Serial Monitor
Message (Enter to send message to 'Arduino Uno' on 'COM4')
Light Sensor Value: 933 | LED Status: ON
Light Sensor Value: 935 | LED Status: ON
Light Sensor Value: 935 | LED Status: ON
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Light Sensor Value: 935 | LED Status: ON

